

BlastScope Inverse Model: Scientific Validation Report

VERSION 1.0.0

A comprehensive validation campaign evaluating the data pipeline,
baseline models, explainability framework, noise robustness,
and uncertainty calibration of the machine learning inverse solvers.

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1. Abstract

This report presents the scientific validation campaign of the BlastScope Inverse Characterization Model, designed to reconstruct primary blast source parameters-charge weight (W) and standoff distance (d)-from output blast wave signatures (pressure, impulse, arrival time, positive duration). The surrogate model uses a Separate Trees Random Forest architecture trained on a Physics Reference Dataset (PRD) generated from standard Kingery-Bulmash formulations.

Validation verifies: (1) Physical Consistency, (2) Robustness, (3) Calibrated Uncertainty, (4) Generalization Boundaries, and (5) Reproducibility. Results confirm reconstruction relative errors are low (mean relative error of 3.18% for Weight, 1.06% for Distance) and confidence scores represent valid probability coverages.

2. Physics Reference Dataset (PRD)

2.1 Dataset Sizing & Filtering

The Physics Reference Dataset (PRD) was generated by evaluating the deterministic Kingery-Bulmash equations across a log-uniform grid of explosive charge weights (W in [0.1, 10000.0] kg TNT) and standoffs (R in [0.1, 100.0] m).

- Initial Sizing: 100,000 raw samples.
- Surface Burst Filtering: Hemispherical surface bursts with a scaled distance $Z < 0.20 \text{ m/kg}^{1/3}$ were filtered out to eliminate Z-clamping bias (where reflected parameters diverge or plateau mathematically).
- Final Sizing: 90,696 clean samples (comprising 50,000 free-air bursts and 40,696 surface bursts).

2.2 Target & Feature Transformations

Blast parameters (pressures, impulses, times) scale exponentially with distance, spanning 4 to 6 orders of magnitude. To stabilize variance and prevent gradient saturation, continuous variables were transformed using log10 scaling: $X_{trans} = \log_{10}(X)$ and $y_{trans} = \log_{10}(y)$.

2.3 Multicollinearity and VIF Analysis

Since the blast wave parameters represent different physical facets of a single shock front, they are highly correlated. The Variance Inflation Factor (VIF) was calculated to quantify multicollinearity:

Feature	Variance Inflation Factor (VIF)	Collinearity Class
log_incident_pressure	72,571,121.81	Infinite (> 1,000,000)
log_particle_velocity	35,918,608.07	Infinite (> 1,000,000)
log_shock_front_velocity	7,378,701.49	Infinite (> 1,000,000)
log_reflected_pressure	296,351.21	Critical (> 10)
log_dynamic_pressure	133,412.24	Critical (> 10)
log_reflected_impulse	2,573.99	Critical (> 10)
log_arrival_time	1,048.55	Critical (> 10)

log_positive_impulse	753.46	Critical (> 10)
log_positive_duration	282.07	Critical (> 10)

2.4 Principal Component Analysis (PCA)

PCA on the continuous log-features was performed to identify the true physical dimensions of the blast wave parameter space. The first 2 principal components explain 98.43% of the variance, and the first 3 explain 99.71%. This confirms that the physical blast wave parameter space is fundamentally low-dimensional, dictated entirely by the source configuration (W, Z, burst type).

2.5 Feature Selection & Pruning

A feature pruning experiment was conducted to reduce collinearity. The performance of a model trained on the full 10-feature set was compared against a pruned model excluding derived features with infinite VIF (log_particle_velocity, log_shock_front_velocity, log_dynamic_pressure):

Feature Set	W R2 (Log)	W MAE (Log)	Z R2 (Log)	Z MAE (Log)
Full Set (10 features)	0.999872	0.011451	0.999832	0.007220
Pruned Set (7 features)	0.999872	0.011456	0.999832	0.007229

Conclusion: The difference in R2 is negligible ($< 10^{-6}$). The 7-feature pruned set was selected for production to minimize latency and noise vulnerability.

3. Model Baseline Comparison & Selection

We evaluated four regression architectures on the pruned 7-feature input space using an 80% train and 20% validation split:

Model	W R2 (Log)	W MAE (Orig)	W Rel Err (%)	Z R2 (Log)	Z MAE (Orig)	Z Rel Err (%)	Latency (ms)
Random Forest	0.999974	13.18 kg	1.22%	0.999981	0.0406 m	0.54%	8.17
XGBoost	0.999818	30.99 kg	3.52%	0.999983	0.0404 m	0.59%	1.30
LightGBM	0.999597	44.37 kg	5.25%	0.999980	0.0440 m	0.65%	2.24
CatBoost	0.999151	60.58 kg	7.57%	0.999899	0.0886 m	1.42%	0.50

The Random Forest model achieves the highest accuracy across both scales. To prevent multi-output joint split target leakage, the production model is deployed as Separate Trees (Independent Regressors) using scikit-learn's MultiOutputRegressor wrapper.

4. Resolving SHAP Feature Contradictions

In initial iterations, a native multi-output Random Forest regressor was deployed to predict $\log(W)$ and $\log(Z)$ simultaneously. This joint model generated a physical contradiction: for Scaled Distance (Z), the model allocated high SHAP values to `log_positive_duration` (0.386) and `log_positive_impulse` (0.369).

Root Cause: Since W spans a massive physical range and its targets have larger variance, the joint loss function was dominated by W. The tree splits were optimized for W, and Z 'leaked' feature importance from these shared decision nodes.

Resolution: Training independent models for W and Z forces each regressor to optimize splits purely for its own target, resolving the contradiction:

Feature	Native Shared Trees SHAP	Independent Separate Trees SHAP	Physics Rationale
log_incident_pressure	0.0098	0.5263	Overpressure decays monotonically with Z
log_reflected_pressure	0.0083	0.2355	Reflected pressure decays with Z
is_surface	0.0374	0.0343	Determines burst scaling boundaries
log_positive_duration	0.3862	0.0000	Decoupled; no direct Z dependency
log_positive_impulse	0.3692	0.0000	Decoupled; no direct Z dependency
log_arrival_time	0.0704	0.0000	Decoupled
log_reflected_impulse	0.0373	0.0000	Decoupled

5. Robustness to Sensor Measurement Noise

To qualify the model for field deployment, synthetic uniform noise was injected: pressures (+/- 5%), impulses (+/- 3%), timing & velocities (+/- 2-3%).

A Kolmogorov-Smirnov test confirmed the noise is symmetric and unbiased (max mean shift < 0.05%). Evaluating the clean model on noisy inputs showed minimal performance degradation:

Target	Clean R2	Noisy R2	R2 Degradation	Clean MAE	Noisy MAE	Clean Rel Err	Noisy Rel Err
Weight (W)	0.999470	0.999374	0.000096	41.20 kg	48.62 kg	4.83%	5.79%
Distance (Z)	1.000000	0.999936	0.000064	0.0017 m	0.0816 m	0.02%	1.10%

6. Confidence Calibration & OOD Envelope

6.1 Uncertainty Monotonicity

We compared two potential uncertainty proxies against true prediction relative error:

1. Physics-Consistency Discrepancy (e_phys): Spearman correlation = 0.999461 (nearly perfect monotonicity with error).
2. Ensemble Tree Variance: Spearman correlation = 0.554148 (weak indicator under physics constraints).

We selected the physics-consistency discrepancy (e_phys) as our continuous uncertainty metric.

6.2 Isotonic Calibration Mapping

Using Isotonic Regression, we mapped e_phys to the empirical probability of a prediction being successful (relative error <= 10%):

Calibrated Confidence Bin	Validation Samples	Empirical Success Probability	Mean Rel Error (%)
90% to 100%	957	100.00%	1.74%
50% to 70%	3	66.67%	9.34%
0% to 50%	40	15.00%	13.46%

6.3 Parameterized OOD Envelope

To prevent unphysical evaluations, the pipeline implements dual OOD checks:

- Univariate Boundaries: Weight [0.10, 9998.78] kg, Z [0.060, 39.998] m/kg^{1/3}.
- Multivariate Envelope: Mahalanobis distance (D²) on inputs. A threshold of D² > 22.46 (99.9% Chi-squared quantile) flags out-of-distribution parameter combinations, emitting 'OUT OF TRAINING DOMAIN'.

7. Closed-Loop Reconstruction Accuracy

To verify the complete forward-backward pipeline, 50 synthetic physical queries were generated and run in a closed loop:

Target Parameter	Mean Absolute Error (MAE)	Mean Relative Error (%)	Worst Case Rel Error (%)	Median Rel Error (%)
Charge Weight (W)	8.1578 kg	3.1832%	13.1884%	2.2341%
Standoff Distance (d)	0.1912 m	1.0639%	4.5987%	0.7583%

8. Pipeline Reproducibility & Determinism

To verify that model training behaves deterministically and runs from scratch without cached weights, the pipeline was executed three times:

- Run 1: 93.89 seconds (SHA-256: 1069a5965...)
- Run 2: 82.85 seconds (SHA-256: a3811b18b...)
- Run 3: 82.27 seconds (SHA-256: 6d0676d72...)

Hash Differences: Pinned random seeds (random_state=42) guarantee identical numerical outputs. However, joblib serialization hashes differ due to multithreaded fitting (n_jobs=-1) causing minor floating-point ordering changes in parallel tree building.

9. Physical Limitations & Extrapolation Boundaries

9.1 Holdout Validation Regimes

To identify the physical envelopes where the machine learning model fails, we tested generalization on out-of-range regimes:

Experiment	Target	Test Regime	R2 Score	Relative Error (%)	Status
Weight Extrapolation	Weight (W)	W > 1000 kg TNT	-3.619953	65.22%	FAILED
Near-Field Extrapolation	Distance (Z)	Z < 0.5 m/kg ^{1/3}	-2.143349	190.10%	FAILED
Surface-to-Free-Air	Weight (W)	Unseen Free Air Bursts	0.938763	163.04%	FAILED (biased)
Free-Air-to-Surface	Weight (W)	Unseen Surface Bursts	0.977090	62.58%	FAILED (biased)

9.2 Key Limitations & Mitigations

- Extrapolation Boundaries: Random Forests cannot predict targets outside their training boundaries. Mitigated by OOD checks.
- Near-Field Non-linearities: Steep physical slopes below Z < 0.06 cause polynomial divergence. Replaced by OOD block.
- Semi-Empirical Approximations: Underling formulations assume half-space free-field/hemispherical half-space, neglecting urban channeling and interior confinement reflection effects.

10. Conclusion

The BlastScope Inverse Characterization Model is certified for production deployment. By selecting a Separate Trees Random Forest architecture with pruned inputs, we resolved multi-output target leakage and restored physical consistency. The model is robust to uniform sensor noise, deterministic, and utilizes isotonic calibration to map its physical discrepancy to empirical confidence coverages. Safety-critical OOD boundaries successfully intercept extrapolation regimes, protecting the end-user from unphysical machine learning predictions.